A-A-54526
23 November 1991

COMMERCIAL ITEM DESCRIPTION

INDICATOR, STERILIZATION

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-I-36789, which is cancelled.

This Commercial Item Description covers one type of sensitized paper or fabric test record for determination of air elimination from within a prevacuum steam sterilizer during the pre-vacuum stage of a high vacuum sterilizer.

Salient Characteristics:

The air removal indicator shall consist of a sensitized sheet of paper having a light background overprinted with a darker colored uniformly applied steam-heat sensitive ink. The ink shall undergo a significant uniform color change in the presence of a vacuum, or absence of air, with subsequent exposure to steam heat during the vacuum portion of the sterilization cycle.

Each air removal indicator sheet shall have minimum dimensions of 8-1/2 inches by 11 inches and shall be sufficiently flexible to conform to the irregular contours in surgical packs. The steam-heat sensitive ink shall be applied in a repetitive or concentric pattern within an area not less than 7 inches by 11 inches and in such a manner that there shall be no area greater than 1/2 inch diameter within the pattern without the indicating ink. Areas with printing that identifies the product or manufacturer shall be excluded from the 1/2 inch diameter requirement. Once the ink has changed, the chemical process shall be irreversible providing a permanent record.

When subjected to a complete vacuum-steam sterilization cycle, at 270°F for 3-1/2 minutes, the indicator's steam-heat sensitive ink shall exhibit a complete and uniform color change as explained in the instructions to indicate complete air removal. When subjected to a sterilization cycle with

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partial vacuum and steam at 270°F or at 250°F, the ink color shall remain unchanged or shall only slightly change to indicate incomplete sterilization.

The following information shall be printed on each indicator sheet:

* Name or registered trademark of the manufacturer
* Product identification or tradename
* For record keeping purposes:

DATE
AUTOCLAVE or STERILIZER NO.
OPERATOR

Full instructions for use shall accompany each unit of thirty-one (31) indicators.

Workmanship. The indicator sheets shall be free from defects which detract from their appearance or impair their serviceability. Sheets shall be free of cuts, tears and holes in the area with the indicating ink pattern and shall be clean.

Unit. Package (PG). One package containing thirty-one indicators with instruction sheet, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the

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supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one indicator sheet.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1 (percent defective) and an inspection level of S-1. The unit of product for test purposes shall be one indicator sheet.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

1. Physical tests. Physical tests shall be conducted in accordance with this specification and applicable methods of TAPPI, as listed below:

Property	TAPPI Method
Thickness	T-411
Tearing resistance	T-414
Porosity	T-460

2. Complete sterilization. A single sheet of the indicator shall be placed in the center of a stack of towels 10 to 11 inches high (2 to 3 dozen towels) and folded flat to 9 by 12 inches. The towels shall be wrapped as an ordinary surgical bundle and autoclaved in a pre-vacuum type sterilizer at 260°F for 12 minutes. Repeat the test, using a fresh indicator, at 272°F for 3-1/2 minutes. At the conclusion of the sterilization cycle, the indicators shall be examined.

3. Incomplete sterilization. The test shall be conducted at the same time and in the same sterilizer as the test in 2. (above). A 9 inch strip cut from one side of the indicator

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paper shall be placed in a Mason jar, or equivalent tight seal container, in such a manner that half of the strip protrudes from the jar. The portion of the strip inside the jar shall be taped to the inner wall so that the temperature outside the jar reaches the strip while air is trapped in the jar. Before sealing, the jar and strip shall be heated in a 250°F oven for 5 minutes in order to eliminate moisture from the sample and jar. The jar shall be tightly sealed, placed in the sterilizer and the two sterilization cycles completed as specified in 2. (above). At the conclusion of each test, the test strip shall be examined.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes,

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ordinances and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

6530-00-181-7770

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6530-2257

Location: STER3CID.1BC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54526	2. DOCUMENT DATE (YYMMDD) 23 November 1991

3. DOCUMENT TITLE
INDICATOR, STERILIZATION

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial AC 215-737-2870	(2) AUTOVON AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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